

Sophie Ruehr
Biosphere Sciences & Engineering Division
Carnegie Institution for Science
Stanford, CA USA

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ACADEMIC APPOINTMENTS

Carnegie Institution for Science

2025-	Postdoctoral Fellow in Land-Surface Modeling Biosphere Sciences & Engineering Division	Stanford, CA
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EDUCATION

Ph.D.	University of California Berkeley	Environmental Science, Policy & Management	2025
B.Sc.	Yale University	Geology & Geophysics	2018

RESEARCH INTERESTS

Terrestrial ecosystems, climate solutions, drought resilience, AI for Earth data, proximal & satellite hyperspectral remote sensing, sensor development, sustainable agriculture, ecohydrology, water management, land-atmosphere interactions, stakeholder perspectives & policy implications

PROFESSIONAL EXPERIENCE

Provincetown Independent

2019-2020	Newspaper Staff Reporter	Provincetown, MA USA
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University of the South Pacific

2018-2019	Yale University Huang Fellow	Port Vila, Vanuatu
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Woods Hole Oceanographic Institution

2017	Summer Student Fellow	Woods Hole, MA USA
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Provincetown Banner

2016	Newspaper Staff Reporter	Provincetown, MA USA
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HONORS & AWARDS

Max Plank-Caltech-Carnegie-Columbia MC³ 4 Earth Center Post-Doctoral Fellowship, 2025-present
NASA Future Investigators in Earth and Space Science and Technology Fellowship, 2022-2025
Achievement Rewards for College Scientists Fellowship, 2020-2022
National Science Foundation Graduate Research Fellowship Program Honorable Mention, 2022
New England Newspaper Association, First Place: Science/Technology Reporting, 2021
New England Newspaper Association, First Place: Health Reporting, 2021
Yale University Parker Huang Undergraduate Travel Fellowship, 2018-2019
Yale University Dept. of Geology & Geophysics Hammer Prize, 2018
Yale University Karen Von Damm 1977 Fellowship, 2017
Woods Hole Oceanographic Institution Summer Student Fellowship, 2017

GRANTS & COMPETITIVE FUNDING AWARDS (\$450,000 total)

2025-	Max Plank-Caltech-Carnegie-Columbia MC ³ 4 Earth Center: Postdoctoral Fellowship in Land-Surface Modeling (\$82,500 annually, up to 4 years)
2025-2026	University of California Berkeley Chancellor's Advisory Committee: Improving field safety and inclusion in ESPM (\$54,000)
2025-2026	University of California Berkeley Be Smart About Safety Grant: Field safety through improved emergency communications and training (\$25,000)
2022-2025	NASA FINESST (Future Investigators in NASA Earth and Space Science and Technology): Quantifying ecosystem reliance on groundwater, Co-PI Ruehr (\$150,000)
2022	FLUXNET Early Career Secondment: Quantifying ecosystem reliance on groundwater with eddy covariance and sap flow data (\$7,400)
2020	University of California Berkeley Carol Baird Fieldwork Grant: Using hyperspectral imagery at Sagehen Experimental Forest to quantify ecosystem groundwater use (\$33,000)
2020	Achievement Rewards for College Scientists Graduate Student Fellowship: Graduate funding for 2 years (\$100,000)
2018	Yale University Parker Huang Undergraduate Travel Fellowship: Oral history in Vanuatu to support paleoclimate research (\$36,000)
2017	Yale University Karen Von Damm 1977 Fellowship: In support of senior honors thesis field research in Lanzhou, China (\$5,000)
2017	Woods Hole Oceanographic Institution Summer Student Fellowship: Hurricane paleoclimatology research (\$8,000)

SERVICE***Committees & management***

UC Berkeley Dept. ESPM Field Safety Committee, 2024-2025

UC Berkeley College of Natural Resources LGBTQ+ Coalition, 2020-2023

AmeriFlux Diversity, Equity & Inclusion Committee, 2020-2024

UC Berkeley Dept. ESPM Graduate Diversity Council, 2020-2024

Reviewing

Regular reviewer for Nature Communications, PNAS, Science Advances, One Earth, Geophysical Research Letters, Agricultural & Forest Meteorology, Nature Communications Earth & Environment, Hydrology, Earth's Future, AGU Advances, & Journal of Arid Environments

MENTORING***Current***

Eden Gonzalez, UC Berkeley undergraduate student, 2023-

Jackson Coldiron, UC Santa Barbara master's student, 2024-

Past

Adam Rashid, UC Berkeley undergraduate student (now: MIT PhD student), 2023-2024

Megan Hur, UC Berkeley undergraduate student (now: NASA Goddard research assistant), 2022-2024

Tyler Goldstein, UC Berkeley undergraduate (now: UC Berkeley science communications), 2021-2023

TEACHING***Guest lectures***

- 2025 “Land Surface Modeling” in ESPM 111, Carbon Cycle Dynamics, UC Berkeley
 2024 “Remote Sensing of the Biosphere” in EPS 251, Ecosystem Science, UC Berkeley

Courses

- 2024 Graduate Student Instructor, Ecosystem Science, led by Prof. Dennis Baldocchi

Centers

- 2020-2024 Data Consultant for R, Python, Google Earth Engine, UC Berkeley D-Lab

PUBLICATIONS (see [Google Scholar Profile](#) for updates)***Published***

- Ospina D et al. [70 authors incl. **Ruehr S**]. 2026. Ten new insights in climate science 2026. *Global Sustainability*, 9:e6. 10.1017/sus.2025.10043.
- Pierrat ZA, Gustine RN, Boser A, **Ruehr S**, Lee CM, Reager JT, Cawse-Bicholson K. 2026. Human contributions to evapotranspiration mitigate swings in dry-to-wet year transitions. *Communications Sustainability*, 1, 8. 10.1038/s44458-025-00002-w.
- **Ruehr S**, Bassiouni M, Kang Y, Socolar Y, Magney T, Keenan TF. 2025. Crop diversification improves water-use efficiency and regional water sustainability. *Environmental Research Letters*, 20, 114062. 10.1088/1748-9326/ae15a9.
- **Ruehr S**, Gerlein-Safdi C, Falco N, Seibert P, Chou C, Albert L, Keenan TF. 2024. Quantifying seasonal and diurnal cycles of solar-induced fluorescence. *Geophysical Research Letters*, 51, 14. 10.1029/2023GL107429.
- **Ruehr S**, Keenan TF, Williams C, Zhou Y, Lu X, Bastos A, Canadell P, Prentice IC, Sitch S, Terrer C. 2023. Evidence and attribution of the enhanced land carbon sink. *Nature Reviews Earth & Environment*, 4, 518-534. 10.1038/s43017-023-00456-3.
- **Ruehr S**, Girotto G, Verfaillie J, Baldocchi D, Cabon A, Keenan TF. 2023. Ecosystem groundwater use enhances carbon sinks in a semi-arid oak savanna. *Agricultural & Forest Meteorology*, 342, 109725. 10.1016/j.agrformet.2023.109725.
- Massoud EC, Andrews L, Reichle R, Molod A, Park J, **Ruehr S**, Girotto M. 2022. Seasonal forecasting skill for the High Mountain Asia region in the Goddard Earth Observing System. *Earth System Dynamics*, 14, 147–171. 10.5194/esd-14-147-2023.
- **Ruehr S**. 2021. Beyond the vulnerability/resilience dichotomy: Perceptions of and responses to the climate crisis on Emau, Vanuatu. *Island Studies Journal*, 10.24043/isj.151.
- **Ruehr S**, Lee X, Smith R, Li X, Xu Z, Liu S, Yang X, Zhou Y. 2020. A mechanistic investigation of the oasis effect in the Zhangye cropland in semiarid western China. *Journal of Arid Environments*, 176, 104120. 10.1016/j.jaridenv.2020.104120.
- Espeland M et al. [14 authors incl. **Ruehr S**]. 2015. Ancient Neotropical origin and recent recolonization: Phylogeny, biogeography and diversification of the Riodinidae (Lepidoptera: Papilionoidea). *Molecular Phylogenetic Evolution*, 93, 296-306. 10.1016/j.ympev.2015.08.006.

In review

- **Ruehr S**, Citrini A, Wendland E, Dukes J, Rosa L. Double cropping increases water scarcity and irrigation demand in Brazilian soybean croplands.
- **Ruehr S**, Pierrat Z, Parazoo N, Keenan TF. Harnessing solar-induced fluorescence for agricultural research and management.
- Rao MP et al. [29 authors incl. **Ruehr S**]. Atmospheric aridity decouples carbon assimilation and growth in temperate deciduous oaks.
- Dannenberg M, Barnes M, Biederman J, Johnston M, Meerdink S, **Ruehr S**, Scott R, Smith W, Williams P. Improved estimation of the interannual variability of dryland carbon and water fluxes.

In preparation

- Cabiyo B, **Ruehr S**, Arora T, Nolan CJ, Kueppers L, Field C. The durability of forests in a changing climate. (Target journal: *Nature*).
- **Ruehr S**, Rosa L. Biofuel's potential in United States constrained by future drought. (Target journal: *Nature Energy*).

Publications from consulting

- Rutkove SB et al. [6 authors incl. **Ruehr S**]. 2022. Design and pilot testing of a 26-gauge impedance-electromyography (iEMG) needle in wild type and ALS mice. *Nerve & Muscle*, 65, 6. 10.1002/mus.27551.
- Chin A, **Ruehr S**, Tarulli A, Rutkove SB. 2007. Saline-saturated Balsa Wood as a Testing Medium for Rotational Electrical Impedance Myography. *IFMBE Proceedings*, 17, 272-275. 10.1007/978-3-540-73841-1_72.

PRESENTATIONS***First author oral presentations***

- Multicropping increases water scarcity and irrigation demand in Brazilian croplands, May 2026, European Geophysical Union, Emerging interdisciplinary perspectives in water resilience and risk science: society–ecology–freshwater nexus approaches and socio-meteorology.
- Ecohydrology insights for water resource management in agroecosystems (**invited**), December 2025, American Geophysical Union Fall Conference, Frontiers in Ecohydrology.
- Understanding and supporting resilient ecosystems (**invited**), August 2025, Lawrence Livermore National Lab, Livermore, CA.
- Emerging satellite products unveil cropland water use efficiency trends and drivers in California's Central Valley, December 2024, American Geophysical Fall Meeting, Washington DC, USA.
- Evidence and attribution of the land carbon sink's historic enhancement (**invited**), October 2023, Max-Planck Institute for Biogeochemistry, Jena, Germany.
- Groundwater drought decreases carbon fixation in a semi-arid oak savannah, September 2023, CREAM, Barcelona, Spain.
- Hyperspectral imagery illuminates drivers of solar-induced fluorescence across landscapes, December 2022, American Geophysical Fall Meeting, Washington DC, USA.

- Groundwater drought decreases carbon fixation in a semi-arid oak savannah, December 2022, American Geophysical Fall Meeting, Chicago IL, USA.
- Quantifying ecosystem reliance on groundwater, December 2021, American Geophysical Fall Meeting, New Orleans LA, USA.
- Carbon emissions and offsets: Global and local research (**invited**), August 2021, Distinguished Speaker at ARCS Forward National Speaker Series.
- Incorporating stakeholder feedback in paleoclimatology research, 2019, Woods Hole Oceanographic Institution MA, USA.
- Tracing ancient cyclones: paleoclimate, oral history & climate futures (**invited**), November 2018, University of the South Pacific Emalus Campus, Vanuatu.
- The oasis effect: evaluating intrinsic biophysical mechanism theory and its implications for sustainable water management in Zhangye, Gansu, China, May 2018, Yale University CT, USA.
- Intrinsic Biophysical Mechanism Theory & the Oasis Effect, March 2018, Key Laboratory of West China's Environmental System, Lanzhou University, Gansu, China.
- Exploring the Oasis Effect with Land Surface Modeling (**invited**), March 2018, School of Geography, Beijing Normal University, Beijing, China.

First author posters

- Evidence and attribution of the land carbon sink's historic enhancement, December 2023, American Geophysical Union Fall Meeting, Chicago IL, USA.
- Picturing SIF: field readiness and initial results from a novel SIF imaging instrument, December 2021, American Geophysical Fall Meeting, New Orleans LA, USA.
- Latent heat drives cooling over oases, December 2020, American Geophysical Fall Meeting.
- Newfound aspects of ancient hurricanes: reconstructing storm intensity and sediment deposition dynamics in northeastern coastal ponds, August 2017, Woods Hole Oceanographic Institution, MA, USA.

MEDIA & OTUREACH

Science communication

2023	FLUXNET blog
2022	Keenan Group TikTok
2022	AmeriFlux 25 years data visualization tool
2022	Berkeley Science Review
2019-2020	Provincetown Independent
2019	InsideClimate News
2010	WOMR Cape Cod's Outermost Radio
2016, 2018	Provincetown Banner

Press

2023	Ask MIT Climate
2023	Phys.org

WORKSHOPS

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| 2025 | Identity-Based Risks in Field Work, UC Berkeley, CA |
| 2024 | Center for Climate Sciences Summer School, NASA Jet Propulsion Lab, Pasadena, CA |
| 2024 | Spring Teaching Conference, UC Berkeley, CA |
| 2024 | FieldFutures Harassment Prevention Training, UC Berkeley, CA |
| 2024 | DroneCamp, CSU Monterey Bay, CA |
| 2022 | Field Safety Workshop, AmeriFlux |
| 2022 | FluxCourse, AmeriFlux at Niwot Ridge Long-term Ecological Research Program, CO |